AI Predicts, ArcGIS Pro Checks: Testing Planning Forecasts Against Reality

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Title	AI Predicts, ArcGIS Pro Checks: Testing Planning Forecasts Against Reality (74/80)
Abstract	AI can forecast redevelopment in seconds, and its maps look convincing. Testing them against reality is harder: public records sit on the street instead of the lot, and buildings finish years after they start. We use ArcGIS Pro as the reality check on a retrospective AI forecast for 42,000 Manhattan lots, made from 2020 data. Joining 2021-2025 certificates of occupancy to tax lots found 430 completed lots where a plain overlay found 16. The AI tied a planning rule of thumb on new-building permits and ranked completions better. Planners and GIS analysts leave with the workflow and a checklist. 599/600 characters. Published as written in the agenda.
Keywords	AI validation; ArcGIS Pro; urban planning; redevelopment forecasting; public records
Topic category	State and local government: Planning
Audience and level	Planners and GIS analysts. Intermediate; no AI background needed.
Primary speaker	Parya Payami, PhD Dissertator, Department of Architecture, University of Wisconsin-Milwaukee
Links	Code and results: github.com/ParyaPayami/proveit Optional, after the talk: the forecasting model as a game, planarena-manhattan.onrender.com/play

Expanded abstract

The problem. AI models now produce redevelopment forecasts in seconds, and the maps look convincing. Testing them against what really happened is where errors hide: public records are placed on the street rather than the lot, buildings are certified years after they start, and each record measures something slightly different from the forecast. A careless test can make a model look better or worse than it is. **The forecasting model is the case study; the ArcGIS Pro workflow is the lesson.**

The case. PlanArena, a parcel-level planning model built as a result of six years of PhD research at the intersection of architecture/urban planning, AI and GIS, gave each of 42,170 Manhattan lots a five-year redevelopment probability. It was trained only on 2015-2020 changes in NYC MapPLUTO and applied to 2020 parcel data; no permits or certificates were used to train it or as inputs. The forecast was made afterwards, so this is a retrospective test, not a live one. The comparison is the planners' soft-site rule (lots built to well under their zoning).

How ArcGIS Pro turned the records into a fair test. The same steps, with one dataset of new-building permits, give the second outcome. An automatic check warned about the street-frontage problem in step 2.

Step in ArcGIS Pro	Tool	Result
1. Get the records	ProveIt download with source notes and known traps	1,772 new-building certificates of occupancy, 2021-2025, in a box around Manhattan; about 600 in Manhattan
2. Plain overlay	Spatial Join (inside)	Only 30 certificates fell inside a lot: **16 completed lots**. Records are placed on the street frontage
3. Join by tax lot	Key join on the tax-lot number (BBL)	547 certificates matched 398 lots
4. Nearest lot for the rest	Generate Near Table, 30 m limit	The rest, mostly condominium billing numbers that are not tax lots, go to the nearest lot: **430 completed lots**. Not yet checked by hand

Step in ArcGIS Pro	Tool	Result
5. Remove projects already underway	Definition query: no building from 2015 or later in 2020	215 of the 430 removed, as their building was likely already started: **215 lots tested**

Results. AUC is the chance that a lot that did redevelop is ranked above one that did not (0.5 is chance). The rule ranks permits correctly, so its low score on certificates is not a reversed scale.

Outcome, 2021-2025 (same lots for both)	AI forecast	Rule of thumb	Difference, 95% range	Read as
New-building permits issued (146 of 41,898 lots)	AUC 0.82	AUC 0.81	-0.01 to 0.04	A tie
New-building certificates of occupancy (210 of 41,399 lots)	AUC 0.67	AUC 0.39	0.25 to 0.33	AI ranks completions better; the rule scores below chance
Same, stricter timing: lots with any building from 2000 or later removed (83 of 39,745 lots)	AUC 0.75	AUC 0.70	0.00 to 0.11	Most of the rule's failure was timing; the AI stays ahead, narrowly

Read honestly. The test checks where, not how many. Permits are not construction starts and certificates are not every completion; both are records with known gaps. Few lots redevelop (about 0.4% received a new-building permit), so even well-ranked lots mostly did not change: the ranking can prioritize review, not predict single lots. The timing filter is a proxy, and agreement with past records is not cause and effect. The value shown is catching evaluation errors before a model's performance is reported, not a change in a city decision.

Beyond this case (supporting material, not in the talk): the same workflow found posted-speed drive times a median 2.2 minutes faster than FDNY's recorded travel times (167 areas), and showed that three Manhattan rezonings allowed 8,423 more homes while 561 were built by 2025. AI coding agents given the workflow passed 32 tasks with known answers (one run each, author-graded).

Fifteen-minute plan. One case, one workflow.

Minutes	Content
0-2	The planning question: which lots will redevelop? The retrospective design and the rule of thumb
2-6	Live in ArcGIS Pro: why a plain overlay finds 16 completed lots and a tax-lot join finds 430
6-10	Timing: removing projects already underway, and how it changes the comparison
10-13	AI against the rule on the same lots; map where the forecast missed
13-15	The workflow and a one-page checklist; QR codes to the code and the game
15-20	Questions

Takeaways. A tested ArcGIS Pro workflow for scoring any prediction layer against later public records, notes on 16 public outcome sources and their traps, and a one-page checklist. Code and results: github.com/ParyaPayami/proveit. Data: NYC MapPLUTO, DOB NOW job filings, DOB certificates of occupancy; PlanArena manuscript (Payami, 2026).